Ride-Hailing System

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Introduction

The rapid growth of ride-hailing services demands efficient and secure platforms. This project presents a ride-hailing platform integrating booking, real-time tracking, payment processing, and ratings & feedback, using **Object-Oriented Analysis and Design** with UML diagrams [1][2]. It also includes dynamic pricing and an emergency **SOS feature**. The system is developed using the Waterfall methodology to ensure reliability and structured development [3].

Methodology

The ride-hailing platform was developed using an **OOAD** methodology with **UML-based** system modeling.
The system architecture and workflow were designed through use case diagrams, sequence diagrams, and relational database modeling to ensure scalability and maintainability.
Core modules include ride booking, driver matching, payment processing, trip tracking, and emergency support functionalities.

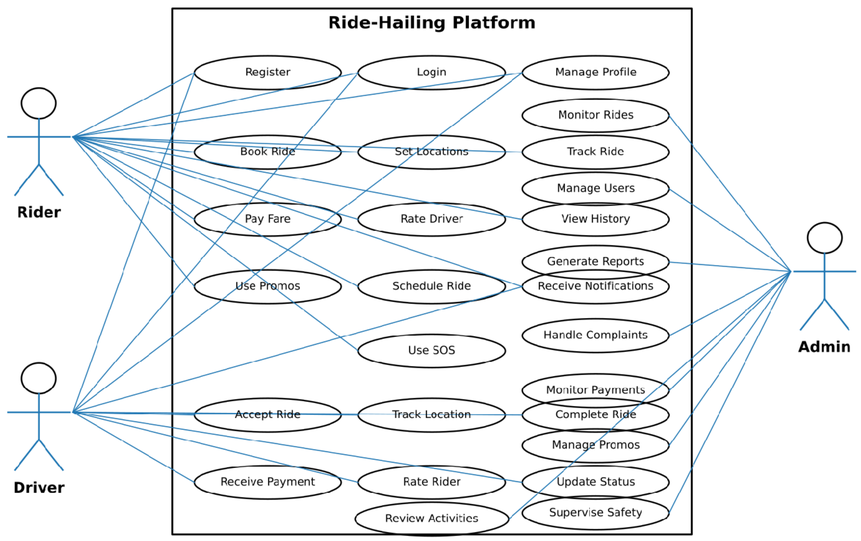


Figure 1: Use Case Diagram (Rider, Driver, Admin)

Results

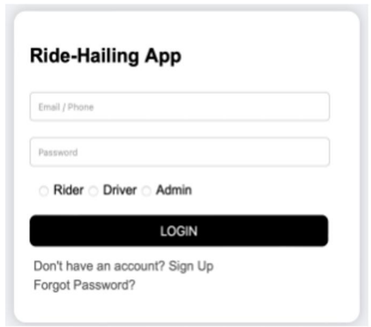


Fig. 1. Login Interface Prototype of the Ride-Hailing Application

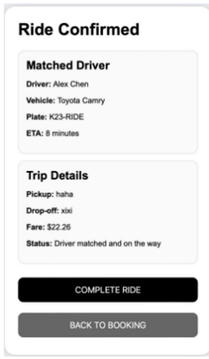


Fig. 3. Ride Confirmation Interface Prototype of the Ride-Hailing Application

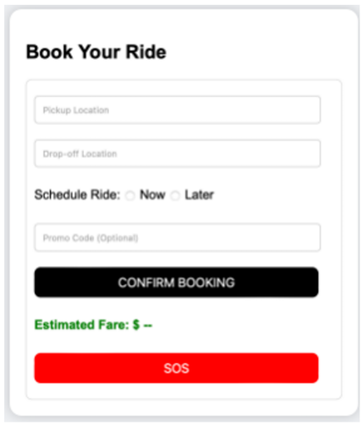


Fig. 2. Ride Booking Interface Prototype of the Ride-Hailin Application

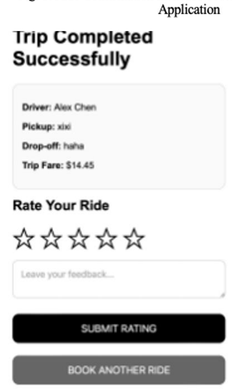


Fig. 4. Trip Completion and Rating Interface of the Ride-Hailing Application

The proposed ride-hailing platform was evaluated through a series of user interface prototypes that demonstrate the complete ride-booking workflow and overall system functionality. The Login Interface provides secure user authentication for both riders and drivers, ensuring safe access to the platform. The Ride Booking Interface enables users to enter pickup and destination locations, select ride options, and submit trip requests efficiently. After a successful driver match, the Ride Confirmation Interface displays trip details, estimated arrival information, and real-time ride status updates. Finally, the Trip Completion and Rating Interface allows users to review completed trips, process payments, and submit feedback through a rating system. These prototypes collectively illustrate the usability, functionality, and **user-centered design** of the proposed ride-hailing application.

Conclusions

This project presents a complete UML-based, object-oriented design of a ride-hailing system using the waterfall model. It covers core features and includes use case, sequence, class, and activity diagrams, plus a seven-table relational database. The implementation follows encapsulation, inheritance, and polymorphism. A testing strategy and cloud deployment plan are provided. Evaluation confirms completeness, UML compliance, and clarity. Future work includes API integration, real-time communication, and distributed architecture for commercial deployment.

References

[1] M. Fowler, UML Distilled: A Brief Guide to the Standard Object Modeling Language, 3rd ed. Boston, MA, USA: Addison-Wesley, 2004.[2] G. Booch, J. Rumbaugh, and I. Jacobson, The Unified Modeling Language User Guide, 2nd ed. Boston, MA, USA: Addison-Wesley, 2005.[3] I. Sommerville, Software Engineering, 10th ed. Boston, MA, USA: Pearson, 2015.[4] M. Martin Fowler, Patterns of Enterprise Application Architecture. Boston, MA, USA: Addison-Wesley, 2002.[5] Uber Technologies Inc., “How Uber Works,” 2025. Available: https://www.uber.com/